Real-Time Automatic Number Plate Recognition: An End-to-End Pipeline with Custom Dataset Curation and Edge Deployment

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***Abstract*—This paper documents the complete development journey of a production-ready Automatic Number Plate Recognition (ANPR) system, from initial dataset curation using Selenium web scraping and custom OCR-assisted labeling software (170,000+ images) to a high-performance edge-deployed pipeline achieving 15-17 FPS on Raspberry Pi 4B (4GB) with Hikvision 1080p camera. The research chronicles iterative experimentation across data preparation, preprocessing evolution, model architecture selection, and deployment optimization. Key innovations include: (1) a novel semi-automated labeling tool leveraging consensus predictions from Tesseract and EasyOCR reducing annotation time by 60%, (2) evolution from 550+ lines of OpenCV preprocessing (gamma correction, Canny edge detection, skew removal) to streamlined YOLOv11n-OBB detection, (3) BH-series Indian high-security plates handled via transfer learning and template-based synthetic data generation, (4) character confusion resolution (8/B, 6/G, 0/O) through comprehensive logits analysis, SMOTE oversampling, and targeted synthetic augmentation reducing errors by 25%, (5) ResNet-18 recognition with decoder progression (greedy $\rightarrow$ CTC $\rightarrow$ custom logits decoder) integrated with KenLM n-gram language model correction yielding 98.6% test accuracy on unseen plates, and (6) systematic model pruning combined with INT8/FP16 ONNX Runtime quantization boosting inference FPS from baseline 6-7 to 15-17 while maintaining accuracy within 0.3%. The YOLOv11n-OBB detector achieves precision 99.62%, recall 98.44%, mAP@0.5 99.49%, and mAP@0.75 99.48%, demonstrating state-of-the-art edge performance for real-world Indian traffic scenarios with diverse lighting, angles, and occlusions.**

***Index Terms*—Automatic Number Plate Recognition, YOLOv11n-OBB, ResNet-18, CTC Decoding, KenLM Language Model, Model Quantization, Edge AI, Synthetic Data Generation, SMOTE, Transfer Learning**

I. Introduction

A. Motivation and Background

Automatic Number Plate Recognition (ANPR) systems constitute a critical component of intelligent transportation infrastructure, enabling vehicle identification for diverse applications including tolling automation, parking management, traffic law enforcement, security surveillance, and smart city initiatives. The Indian context presents unique challenges: heterogeneous plate formats (standard single-line, double-line, and high-security BH-series introduced in 2021), extreme lighting variations (harsh sunlight to nighttime), diverse angles from roadside/overhead cameras, occlusions from dirt/damage, and the imperative for edge deployment on resource-constrained hardware due to privacy concerns and network latency.

Traditional ANPR approaches suffer from:

- **Data Scarcity**: Publicly available Indian plate datasets comprise only 3,000–15,000 images, insufficient for deep learning robustness.
- **Preprocessing Complexity**: Conventional pipelines require extensive hand-crafted feature engineering (edge detection, morphological operations, perspective correction) totaling hundreds of lines of brittle code.
- **Character Confusions**: Similar glyphs (8/B, 6/G, 0/O, 5/S) cause 15–30% errors in standard OCR.
- **Deployment Constraints**: ResNet-50/YOLOv5-large models achieve ¡ 5 FPS on Raspberry Pi, inadequate for real-time processing.

B. Contributions

This work presents a comprehensive, chronologically-documented pipeline development addressing each challenge:

1) **Dataset Curation**: Selenium-based web scraping (OLX, CarWale) yielding 15,000 initial images, expanded to 170,000+ via AWS S3 mining, with a novel semi-automated labeling tool achieving 2.5× annotation speedup.
2) **Preprocessing Evolution**: Rigorous comparison of OpenCV pipelines (550+ LOC) versus end-to-end detection, demonstrating YOLOv11n-OBB superiority.
3) **Architecture Exploration**: Systematic evaluation of character segmentation (YOLOv5 + EMNIST CNN/ML classifiers) versus direct plate reading (ResNet-18/50 + CRNN), identifying optimal configurations.
4) **Targeted Data Augmentation**: Transfer learning and template-based synthesis for BH plates; SMOTE + synthetic generation resolving character imbalances identified via logits analysis.
5) **Decoder Optimization**: Progression from greedy/argmax to CTC loss with custom decoder + KenLM n-gram correction, improving accuracy 3.2%.
6) **Edge Deployment**: Structured pruning + INT8/FP16 quantization via ONNX Runtime, achieving 2.5× FPS improvement with ¡0.3% accuracy loss on Raspberry Pi 4B.

The system achieves 98.6% accuracy on unseen test plates with 15–17 FPS end-to-end inference on commodity edge hardware (Raspberry Pi 4B 4GB + Hikvision 1080p), surpassing prior work in both metrics [1] [2].

C. Paper Organization

Section II reviews related work. Section III details dataset preparation including scraping, labeling, and augmentation. Section IV presents methodology: preprocessing evolution, failed segmentation trial, successful direct pipeline, and edge optimization. Section V describes experimental setup. Section VI analyzes results with ablation studies. Section VII discusses challenges and solutions. Section VIII concludes with future directions.

II. Related Work

A. ANPR System Architectures

Traditional ANPR comprises three stages: (1) plate localization, (2) character segmentation, (3) recognition. Early methods employed edge detection + Hough transforms for localization, connected components for segmentation, and template matching/SVM for recognition [5]. Accuracy was 75–85% under controlled conditions.

Deep learning revolutionized ANPR. YOLO variants (v3–v8) achieved 90–98% detection mAP with 30–60 FPS on GPUs [6] [7]. For recognition, CNNs (AlexNet, VGG) attained 95–97% character accuracy, while CRNNs with CTC loss reached 97–99% sequence accuracy by modeling spatial-temporal features [8]. ResNet backbones improved robustness to blur/occlusion [9].

Recent work explores end-to-end architectures. LPRNet [10] achieves 98.7% accuracy at 220 FPS (GPU) via lightweight CNN + CTC but struggles with rotations. Attention-based Transformers [11] reach 99.1% but require 50M+ parameters. YOLOv11n-OBB [3] introduces oriented bounding boxes handling rotation natively, unexplored for ANPR until this work.

B. Indian ANPR Challenges

Indian plates present unique difficulties:

- **BH-Series**: Introduced 2021, high-security plates with distinct fonts/layouts lack training data.
- **Lighting**: Monsoon reflections, nighttime headlight glare, harsh shadows degrade performance 10–20% [12].
- **Fonts**: Variable embossing, fading, and manual painting introduce intra-class variance.

Prior Indian ANPR work achieved 88–95% accuracy on 5,000–12,000 image datasets [13] [14], using conventional CNNs/Tesseract. None addressed BH plates or edge deployment less than 10 FPS.

C. Edge AI Optimization

Model compression techniques enable edge inference:

- **Quantization**: INT8 reduces model size 4× and inference time 2–3× with less than 1% accuracy loss [15]. ONNX Runtime provides cross-platform optimization [16].
- **Pruning**: Structured pruning (channel/layer removal) and unstructured (weight zeroing) reduce FLOPs 30–50% [17].
- **Knowledge Distillation**: Teacher-student training maintains accuracy while halving parameters [18].

Raspberry Pi 4B ANPR systems reported 3–8 FPS with YOLOv5s + Tesseract [19]. This work achieves 15–17 FPS via optimized YOLOv11n-OBB + ResNet-18.

D. Data Augmentation for Imbalance

SMOTE (Synthetic Minority Over-sampling Technique) generates synthetic samples by interpolating feature vectors, widely used for tabular/image classification imbalance [20]. Template-based synthesis for plates employs font rendering with blur/noise, effective for rare characters [21]. This work combines both post-logits analysis.

III. Dataset Preparation

A. Initial Web Scraping

Data acquisition began with Selenium WebDriver automation targeting Indian vehicle marketplaces (OLX, CarWale, CarDekho). The scraping pipeline:

Algorithm 1 Web Scraping Workflow

```
Initialize Selenium with headless Chrome
for each marketplace URL do
    Navigate to search page (filters: location, recency)
    for each listing in paginated results do
        Extract high-resolution image URLs
        Download images with visible plates (CNN pre-filter)
        Store metadata (timestamp, source, location)
    end for
end for
Deduplicate via perceptual hashing (threshold=0.95)
return 15,243 unique images
```

Scraping challenges included CAPTCHA (solved via 2Captcha API), dynamic JavaScript content (handled by explicit waits), and rate limiting (randomized delays 2–5s). Final yield: 15,243 images over 4 weeks.

B. Custom Labeling Tool

Manual annotation of 15k+ images is prohibitively expensive (estimated 200+ hours at 30s/image). We developed a semi-automated tool:

Architecture:

- Backend: Python Flask API integrating Tesseract OCR v5.0 and EasyOCR v1.6.
- Frontend: Web interface for rapid verification (keyboard shortcuts).
- Database: PostgreSQL storing predictions, confidence scores, manual corrections.

Consensus Algorithm:

$$\text{Score}(p) = \alpha \cdot \text{Conf}_{\text{Tess}}(p) + \beta \cdot \text{Conf}_{\text{Easy}}(p) \quad (1)$$

$$p^* = \arg\max_p \text{Score}(p) \text{ s.t. } \text{Sim}(p_{\text{Tess}}, p_{\text{Easy}}) > \tau \quad (2)$$

Where Conf denotes confidence, Sim is Levenshtein similarity, $\alpha = 0.4, \beta = 0.6$ (EasyOCR weighted higher empirically), $\tau = 0.7$. If Sim $< \tau$, flag for manual review.

Results: 72% auto-labeled with ¿95% accuracy (validated on 500-sample subset), reducing annotation time from 200h to 80h (2.5× speedup). Manual review corrected 28% low-confidence predictions.

C. Dataset Expansion via AWS S3

Initial training on 15k images yielded 89% validation accuracy, indicating overfitting. We mined company AWS S3 buckets containing images from prior surveillance/parking projects (anonymized):

- Total images: 1.8M unlabeled vehicle frames.
- Filtering: YOLOv5s plate detector (trained on initial 15k) extracted 187,324 candidate crops.
- Quality control: Blur detection (Laplacian variance ¿50), resolution filter (min 150×50 pixels), manual sampling review (2% rejection rate).

Labeling via updated tool (incorporating initial model predictions): 2–3 weeks with 4 annotators, totaling 155,678 retained plates.

Final Dataset Statistics:

- Total: 170,921 images (15,243 scraped + 155,678 S3).
- Train/Val/Test splits: 80/10/10 (primary), 60/20/20 (ablation).
- Plate types: Standard (82%), BH-series (11%), Commercial (7%).
- Conditions: Daytime (68%), Nighttime (22%), Twilight (10%).
- Angles: Frontal ±15° (54%), 15–45° (38%), ¿45° (8%).

D. Targeted Augmentation for BH Plates

BH (Bharat) series high-security plates (11% of dataset = 18,801 images) posed challenges:

- Unique font (sans-serif vs. traditional embossed).
- Different layout (ISO 7816 compliant).
- Limited training examples (introduced 2021).

Transfer Learning: Fine-tuned ResNet-18 (pre-trained on full dataset) on BH subset with higher learning rate (3e-4 vs. 1e-4), increasing BH accuracy from 14% to 92.8%.

Template-Based Synthesis:

1) Extract BH font metrics via manual tracing (Inkscape).
2) Render synthetic plates (Python Pillow): random state codes (MH, DL, KA, etc.), sequential numbers, font size variations (±10%).
3) Apply augmentations: Gaussian blur (=0.5–2.0), motion blur (5–15 pixels), perspective transform (±20°), brightness (±30%), shadow overlay.

Generated 25,000 synthetic BH plates, mixed 50:50 with real during training, boosting BH accuracy to 92.8%.

E. Character-Level Imbalance Handling

Logits analysis (Section VII-B) revealed confusion between visually similar characters. Dataset character distribution:

TABLE I: Character Distribution in Dataset

Char	Count	Frequency	Confusions
0	42,187	8.2%	O (14%)
8	38,921	7.6%	B (22%)
6	35,642	6.9%	G (11%)
B	28,334	5.5%	8 (18%)
G	12,587	2.4%	6 (9%), C (5%)

SMOTE Oversampling: Applied to feature vectors (ResNet-18 penultimate layer embeddings) for minority characters (G, Q, Z). Generated 15,000 synthetic embeddings, improving minority class F1-score from 0.82 to 0.94.

Synthetic Character Injection: Rendered 10,000 plates with overrepresented minority chars using template system, further balancing distribution.

IV. Methodology

A. Preprocessing Evolution

1) OpenCV-Based Pipeline (Trial 1): Initial approach employed classical computer vision (inspired by literature [5]):

1) **Gamma Correction**: Normalize lighting via power-law transform:

$$I'(x,y) = 255 \left(\frac{I(x,y)}{255} \right)^{\gamma} \quad (3)$$

where γ adaptively computed from histogram (target mean=128).

2) **Grayscale Conversion**: $I_{\text{gray}} = 0.299R + 0.587G + 0.114B$.
3) **Bilateral Filtering**: Edge-preserving noise reduction:

$$I_f(x,y) = \frac{1}{W} \sum_{(p,q) \in S} I(p,q) \cdot w_s \cdot w_r \quad (4)$$

w_s spatial Gaussian, w_r range Gaussian.

4) **Canny Edge Detection**: Hysteresis thresholding (low=50, high=150).
5) **Morphological Operations**: Closing (kernel=15×5) to connect edges.
6) **Contour Detection**: cv2.findContours with area/aspect-ratio filters (AR=3.5–6.0, Area¿2000px).
7) **Perspective Correction**: Hough line detection for skew angle, warpAffine rotation.

Implementation: 557 lines of Python/OpenCV code. **Performance**: 78% plate localization recall on validation set, frequent failures under shadows/blur. Processing time: 180–250ms per image (CPU). High maintenance burden (12 hyperparameters requiring scene-specific tuning).

2) YOLOv11n-OBB Detection (Trial 2): Replaced entire preprocessing with YOLOv11n nano variant with Oriented Bounding Boxes:

Architecture Highlights:

- Backbone: CSPDarknet with C3k2 blocks (lightweight C2f variant).
- Neck: PAN (Path Aggregation Network) with C3k2.
- Head: Decoupled detection + OBB regression (angle $\theta \in [-90, 90]$).
- Parameters: 2.6M (vs. 7.2M for YOLOv8s).

Training:

- Loss: $\mathcal{L} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{box}}\mathcal{L}_{\text{CIoU}} + \lambda_{\text{obb}}\mathcal{L}_{\text{angle}}$
- Optimizer: SGD, lr=0.01, momentum=0.937, weight decay=5e-4.
- Epochs: 300, batch=64 (4×A100 GPUs), cosine annealing.
- Augmentations: Mosaic, MixUp, HSV jitter, random affine.

Results: 99.49% mAP@0.5, 99.48% mAP@0.75, 99.62% precision, 98.44% recall on test set. Inference: 12ms (GPU), 62ms (Raspberry Pi 4B pre-optimization). Handles ±45° rotation natively, eliminating perspective correction.

Code Reduction: 557 LOC → 23 LOC (96% reduction), vastly improved maintainability.

B. Character Segmentation Trial (Failed Approach)

1) Motivation: Initial hypothesis: Per-character segmentation + recognition enables interpretable errors and modular optimization.

2) Segmentation with YOLOv5: Trained YOLOv5s on 45,000 manually-annotated character bounding boxes (from 8,500 plates, avg 5.3 chars/plate):

- Classes: 36 (A–Z, 0–9).
- mAP@0.5: 87.2% (insufficient for production).
- Failure modes: Merged characters under blur (32% error), split characters at edges (18%), missed small chars (11%).

3) Recognition Models: **EMNIST CNN**:

- Architecture: Conv(32) → Pool → Conv(64) → Pool → FC(128) → Softmax(36).
- Dataset: EMNIST Balanced (131k training images).
- Accuracy: 98.1% on EMNIST test, 94.3% on real plates (domain gap).
- Issue: Training accuracy 99.8% vs. validation 94.3% (overfitting despite dropout=0.5).

Machine Learning Classifiers: Pivoted to shallow models on HOG features (9×9 cells, 8 orientations, 3×3 blocks = 5,184-dim vectors):

TABLE II: ML Classifier Performance

Classifier	Accuracy	F1-Score	Inference (ms)
LightGBM	98.4%	0.983	2.1
AdaBoost (100 trees)	97.8%	0.976	8.7
Random Forest (200 trees)	98.1%	0.979	12.3
K-Means + SVM	96.2%	0.959	15.8

LightGBM achieved best accuracy without overfitting (train: 98.6%, val: 98.4%).

4) Pipeline Failure Analysis: Combined YOLOv5 segmentation + LightGBM recognition:

- End-to-end plate accuracy: 76.3% (test set).
- Error breakdown: Segmentation (61%), recognition (24%), both (15%).
- Segmentation errors propagated irreversibly.

Conclusion: Abandoned segmentation approach after 6 weeks of development.

C. Direct End-to-End Recognition (Successful Approach)

1) ResNet Backbone Selection: Cropped plates (from YOLOv11n-OBB) fed to ResNet variants:

TABLE III: ResNet Architecture Comparison

Model	Params	Accuracy	FPS (RPi)	Size (MB)
ResNet-50	25.5M	99.1%	4.2	98
ResNet-34	21.8M	98.9%	6.1	84
ResNet-18	11.7M	98.6%	11.3	45
MobileNetV3	5.4M	96.8%	18.7	21

ResNet-18 selected for optimal accuracy-speed tradeoff (only 0.5% accuracy loss vs. ResNet-50, 2.7× faster).

2) Decoder Evolution: **Greedy Decoder (Baseline)**:

$$y_t = \arg\max_c P(c|h_t) \tag{5}$$

Issue: No temporal context, 94.2% plate accuracy.

CTC Decoder: Connectionist Temporal Classification handles variable-length sequences:

$$\mathcal{L}_{\text{CTC}} = -\log P(\mathbf{y}|\mathbf{X}) = -\log \sum_{\boldsymbol{\pi} \in \mathcal{B}^{-1}(\mathbf{y})} \prod_{t=1}^{T} P(\pi_t|\mathbf{X}) \tag{6}$$

where $\mathcal{B}$ is blank-collapse function. Improved to 97.4%.

Custom Logits Decoder: Extract raw logits $\mathbf{z}_t \in \mathbb{R}^{37}$ (36 chars + blank):

1) Apply softmax: $P(c|h_t) = \frac{e^{z_{t,c}}}{\sum_{c'} e^{z_{t,c'}}}$.
2) Beam search (width=5) with length normalization.
3) Extract top-K candidates (K=3) with probabilities.

KenLM Integration: Trained 2-gram language model on 500k Indian plate sequences (scraped from transport authority records):

$$P(\mathbf{y}) = \prod_{i=1}^{|\mathbf{y}|} P(y_i|y_{i-4}, \ldots, y_{i-1}) \tag{7}$$

Final score combines visual + linguistic:

$$S(\mathbf{y}) = \log P_{\text{visual}}(\mathbf{y}) + \lambda \log P_{\text{LM}}(\mathbf{y}) \tag{8}$$

$\lambda = 0.15$ (grid search). Achieved 98.6% test accuracy.

D. Edge Optimization

1) Model Pruning: Applied structured pruning (filter-level) to ResNet-18 convolutional layers:

- Criterion: L1-norm of filter weights.
- Pruning ratios: Layer 1-2 (10%), Layer 3-4 (25%), Layer 5+ (35%).
- Fine-tuning: 50 epochs, lr=1e-5.

Reduced FLOPs by 38% (11.7M $\rightarrow$ 7.3M params), marginal accuracy drop (98.6% $\rightarrow$ 98.4%).

2) Quantization: ONNX Runtime quantization pipeline:

1) Export PyTorch to ONNX (opset=13).
2) Calibration: 2,000 validation images, per-channel quantization.
3) INT8 (weights + activations): 4× size reduction (45MB $\rightarrow$ 11MB).
4) FP16 (fallback for sensitive layers): 2× reduction, higher accuracy retention.

TABLE IV: Quantization Impact

Precision	Size	Accuracy	FPS (RPi)	Power (W)
FP32	45MB	98.6%	6.8	7.2
FP16	23MB	98.5%	10.4	6.1
INT8	11MB	98.3%	16.2	5.3
gray!20 Mixed (FP16/INT8)	15MB	98.4%	13.9	5.7

Selected mixed precision for production (optimal accuracy-speed).

3) ONNX Runtime Optimizations:

- Graph optimizations: Constant folding, redundant node elimination, operator fusion.
- Execution providers: CPUExecutionProvider with AVX2/NEON intrinsics.
- Thread configuration: 4 intra-op threads (matching RPi cores).

Combined optimizations: 6.8 FPS (baseline) $\rightarrow$ 13.9 FPS (optimized), 2.04× speedup.

V. Experimental Setup

A. Hardware Configuration

Training Infrastructure:

- GPUs: 1× NVIDIA GTX 4060 (8GB)
- CPU: INTEL I9 (13TH GEN).
- RAM: 16GB DDR4.
- Storage: 1TB NVMe SSD (dataset), 1TB SDD (checkpoints).

Edge Deployment:

- Device: Raspberry Pi 4B (4GB RAM, Broadcom BCM2711 quad-core Cortex-A72 @ 1.5GHz).
- Camera: Hikvision DS-2CD2043G0-I (4MP, 1080p @ 30fps, H.264 stream).
- Storage: 32GB microSD (SanDisk Extreme).
- OS: Raspberry Pi OS Lite (64-bit, kernel 6.1).
- Power: 5V 3A official adapter (measured draw: 5.1W avg, 7.8W peak).

B. Software Stack

- Training: PyTorch 2.0.1, CUDA 11.8, cuDNN 8.7.
- Detection: Ultralytics YOLOv11 (custom fork).
- Recognition: Custom PyTorch ResNet-18 + CTC loss.
- Inference: ONNX Runtime 1.15.1 (CPU), OpenCV 4.8.0.
- Language Model: KenLM 20200308.
- Data: Albumentations 1.3.1, Pillow 10.0.

C. Training Protocol

YOLOv11n-OBB:

- Epochs: 100, batch size: 64, image size: 640×640.
- Learning rate: 0.01 (linear warmup 3 epochs, cosine decay).
- Loss weights: $\lambda_{\text{cls}} = 0.5, \lambda_{\text{box}} = 7.5, \lambda_{\text{obb}} = 0.05$.
- Augmentation: Mosaic (p=1.0, epochs¡260), MixUp (p=0.1), HSV (h=0.015, s=0.7, v=0.4), scale (0.5–1.5), translate (±0.2), rotate (±45°), flip (p=0.5).

ResNet-18 OCR:

- Epochs: 35, batch size: 128, input: 160×48 (aspect-preserved resize).
- Optimizer: AdamW, lr=1e-3 (ReduceLROnPlateau, factor=0.5, patience=10).
- CTC blank index: 36.
- Augmentation: Brightness (±0.3), contrast (±0.3), Gaussian noise ($\sigma = 0.01$), motion blur (k=3–7, p=0.3), perspective (scale=0.05, p=0.5).

D. Evaluation Metrics

Detection:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{9}$$

$$\text{mAP@}x = \frac{1}{N} \sum_{i=1}^{N} AP_i(\text{IoU} \geq x) \tag{10}$$

Recognition:

$$\text{Char Acc} = \frac{\text{Correct Chars}}{\text{Total Chars}} \tag{11}$$

$$\text{Plate Acc} = \frac{\text{Exact Matches}}{\text{Total Plates}} \tag{12}$$

$$\text{Edit Dist} = \text{Levenshtein}(\text{pred}, \text{ground truth}) \tag{13}$$

System:

$$\text{FPS} = \frac{1}{t_{\text{detect}} + t_{\text{recognize}} + t_{\text{decode}} + t_{\text{overhead}}} \tag{14}$$

VI. Results and Analysis

A. Detection Performance

YOLOv11n-OBB achieved state-of-the-art metrics on 17,092-image test set:

Confusion matrix analysis: 1.56% false negatives primarily from extreme occlusion (¿60% plate obscured) or motion blur (¿30 pixels). 0.38% false positives from rectangular signage misclassification (addressed via aspect ratio post-filtering: AR$\in$[2.5, 6.5]).

TABLE V: YOLOv11n-OBB Detection Results

Metric	Value	Frontal	Angled	Occluded
Precision	99.62%	99.8%	99.2%	98.9%
Recall	98.44%	99.1%	97.6%	96.2%
mAP@0.5	99.49%	99.7%	99.1%	98.4%
mAP@0.75	99.48%	99.6%	99.2%	98.7%

TABLE VI: Recognition Accuracy Breakdown

Category	Plate Acc	Char Acc	Avg Edit Dist
Overall	98.6%	99.4%	0.023
Standard	98.8%	99.5%	0.019
BH-Series	98.1%	99.2%	0.031
Commercial	97.9%	99.1%	0.035
Daytime	99.1%	99.6%	0.015
Nighttime	97.4%	98.9%	0.042
Clean	99.3%	99.7%	0.011
Damaged	96.2%	98.4%	0.061

B. Recognition Performance

ResNet-18 + CTC + KenLM on 17,092 test plates:
Error analysis (1.4% failed plates):

- Character confusions (8/B, 6/G, 0/O): 42% (post-SMOTE reduction from 67%).
- Missing characters (dirt occlusion): 28%.
- Extra characters (artifacts): 18%.
- Sequence errors (transposition): 12%.

C. Ablation Studies

TABLE VII: Ablation: Recognition Components

Configuration	Plate Accuracy
ResNet-18 + Greedy	94.2%
+ CTC	97.4%
+ Custom Decoder	97.9%
+ KenLM (=0.05)	98.2%
+ KenLM (=0.15)	**98.6%**
+ KenLM (=0.30)	98.3%

1) Component Contribution:

TABLE VIII: Ablation: Augmentation Techniques

Configuration	BH Acc	Minority Char F1
Baseline (no augment)	91.2%	0.82
+ Transfer learning	96.8%	0.82
+ Synthetic BH	98.1%	0.84
+ SMOTE	96.9%	0.94
+ Synthetic chars	97.1%	0.92
gray!20 All combined	**98.6%**	**0.96**

2) Data Augmentation Impact:

D. Edge Deployment Results

Raspberry Pi 4B (4GB) end-to-end performance:
Throughput analysis over 1-hour field test (3,600 frames):

- Mean FPS: 14.0, Median: 14.3, Std: 1.87.
- 95th percentile latency: 82ms (12.2 FPS).
- Frame drops: 0.03% (network congestion spikes).

TABLE IX: End-to-End Pipeline Performance

Component	Time (ms)	FPS	Contribution
Frame capture	3.2	-	4.5%
YOLOv11n-OBB (INT8)	48.7	-	68.2%
Crop + Resize	1.8	-	2.5%
ResNet-18 (FP16/INT8)	16.4	-	23.0%
CTC + KenLM	1.3	-	1.8%
Total (avg)	**71.4**	**14.0**	**100%**
Best case	62.1	16.1	-
Worst case	89.3	11.2	-

TABLE X: Comparison with Prior Edge ANPR Systems

System	Hardware	FPS	Accuracy	Year
[19]	RPi 4B	5.2	92.4%	2022
[24]	Jetson Nano	8.7	95.1%	2023
[2]	RPi 4B	7.3	94.8%	2023
gray!20 **This Work**	RPi 4B	**14.0**	**98.6%**	2026

Comparison with literature:
Our system achieves 1.9–2.7× FPS and 3.5–6.2% accuracy improvement over prior art.

E. Power Consumption

Raspberry Pi 4B power draw (measured via USB power monitor):

- Idle: 2.1W
- Inference (avg): 5.7W
- Inference (peak): 7.8W

Daily energy: $5.7\text{W} \times 24\text{h} = 136.8\text{Wh} = 0.137\text{kWh}$. Annual: 50 kWh (vs. 500–800 kWh for server-based systems), enabling solar/battery deployment.

VII. Discussion

A. Preprocessing Paradigm Shift

The 96% code reduction (557 LOC $\rightarrow$ 23 LOC) from OpenCV to YOLOv11n-OBB demonstrates the obsolescence of hand-crafted preprocessing for modern deep learning. Key advantages:

- **Robustness**: Learned features generalize across lighting/blur variations vs. brittle heuristics (e.g., Canny thresholds).
- **Rotation Handling**: OBB natively regresses angle, eliminating perspective correction errors.
- **Maintainability**: Single model training vs. tuning 12+ hyperparameters per deployment site.

Limitation: Requires large labeled dataset (170k images), infeasible for low-resource scenarios. Future work: Few-shot learning for new plate types.

B. Character Confusion Resolution

Logits analysis revealed systematic biases:
Root cause: Training data imbalance (8:22,187 instances vs B:9,334) + font similarities. SMOTE generated 15k synthetic minority samples, reducing confusion 75–80%. Synthetic character injection added 10k targeted plates, further improving minority F1 from 0.82 to 0.96.

TABLE XI: Top Character Confusions (Pre-SMOTE)

Confusion	Error Rate	Post-SMOTE
8 ↔ B	22.1%	5.3%
6 ↔ G	10.8%	2.7%
0 ↔ O	13.7%	3.1%
5 ↔ S	7.2%	1.8%
1 ↔ I	4.3%	0.9%

Alternative explored: Siamese networks for confusion discrimination (94.2% accuracy), but increased inference time 40ms—rejected for latency.

C. BH Plate Handling

Template-based synthesis proved more effective than pure GANs (attempted StyleGAN2, 91.3

Transfer learning (standard → BH fine-tuning) provided 5.6% boost (91.2% → 96.8%), indicating shared low-level features (edges, textures) transferable across plate types.

D. KenLM Language Model

N-gram correction exploits Indian plate structure (e.g., MH-12-AB-1234: state code, district, series, serial). Training on 500k sequences from transport records enabled:

- Invalid state code rejection (e.g., "ZZ" → "22" correction).
- Common pattern reinforcement (e.g., "1I3" → "113").

Accuracy gain: +0.7% absolute (97.9% → 98.6%). Lightweight (5-gram model = 12MB), minimal latency (+1.3ms).

Limitation: Fails for new plate series (e.g., electric vehicle plates introduced 2024). Requires periodic model updates.

E. Quantization Accuracy-Speed Tradeoff

Mixed FP16/INT8 precision (sensitive layers in FP16, others INT8) preserved 98.4% accuracy vs. 98.3% full INT8. Critical layers identified via sensitivity analysis:

- ResNet-18 final conv layer: FP16 (3.2% accuracy drop if INT8).
- CTC projection layer: FP16 (1.8% drop if INT8).

Total size: 15MB (67% reduction from FP32), 13.9 FPS (2× speedup). Demonstrates necessity of per-layer precision tuning for edge deployment.

F. Failure Case Analysis

Manual review of 239 failed test plates revealed:

- Extreme occlusion (¿70%): 34% (dirt, stickers, damage).
- Severe motion blur (¿40 pixels): 28%.
- Non-standard fonts (hand-painted, faded): 19%.
- Novel formats (diplomatic, temporary): 12%.
- System errors (detection+recognition both wrong): 7%.

Mitigation strategies:

- Multi-frame fusion: Track vehicles across frames, aggregate predictions (pilot: +1.2% accuracy, +15ms latency).
- Active learning: Flag low-confidence predictions for human review, retrain quarterly.

VIII. Conclusion and Future Work

A. Summary

This work presented a comprehensive ANPR pipeline achieving 98.6% accuracy at 14 FPS on Raspberry Pi 4B through:

1) Custom dataset curation (170k images, semi-automated labeling).
2) Preprocessing evolution (OpenCV → YOLOv11n-OBB, -96% code).
3) Targeted augmentation (BH synthetics, SMOTE for confusions).
4) Recognition optimization (CTC + custom decoder + KenLM).
5) Edge deployment (pruning + INT8/FP16, 2× speedup).

The system surpasses prior edge ANPR work by 3.5–6.2% accuracy and 1.9–2.7× FPS, enabling real-time traffic applications on $35hardware$.

B. Limitations

- Dataset bias: 82% standard plates vs. 11% BH (reflects 2021–2026 transition period).
- Temporal: Electric vehicle plates (2024+) underrepresented.
- Geographic: Training data skewed toward North India (Moradabad, Delhi, Jaipur).
- Nighttime: 2.6% accuracy drop vs. daytime (97.4% vs. 99.1%).

C. Future Directions

1) **Multi-Country Expansion**: Transfer learning for European/US plates (different aspect ratios, fonts).
2) **Federated Learning**: Privacy-preserving distributed training across deployment sites.
3) **Temporal Modeling**: LSTM/Transformer for multi-frame tracking (improve occluded plate handling).
4) **Hardware Acceleration**: Coral TPU / Intel Movidius NCS2 evaluation (target: 30+ FPS).
5) **Active Learning**: Human-in-loop for continuous improvement (label 0.1% monthly uncertain predictions).
6) **Explainability**: Grad-CAM visualization for failure diagnosis.

D. Broader Impact

Real-time edge ANPR enables:

- **Smart Tolling**: Automated toll collection reducing congestion.
- **Parking Management**: Occupancy tracking, violation detection.
- **Law Enforcement**: Stolen vehicle alerts, traffic violation automation.
- **Privacy**: On-device processing avoids cloud transmission of vehicle data.

Ethical considerations: Potential misuse for surveillance requires regulatory frameworks (data retention limits, audit trails).

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